

Inequality and Segregation Lab 1

Yukun Jiao

2025-10-27

This report investigates the differences between Sweden and Hungary in their Lorenz curves, Gini coefficients, and the variance decompositions of income, then provides a preliminary analysis of the results. The results indicate that Sweden's income distribution is more equal. The variance decomposition further highlights two distinct patterns.

Part 1

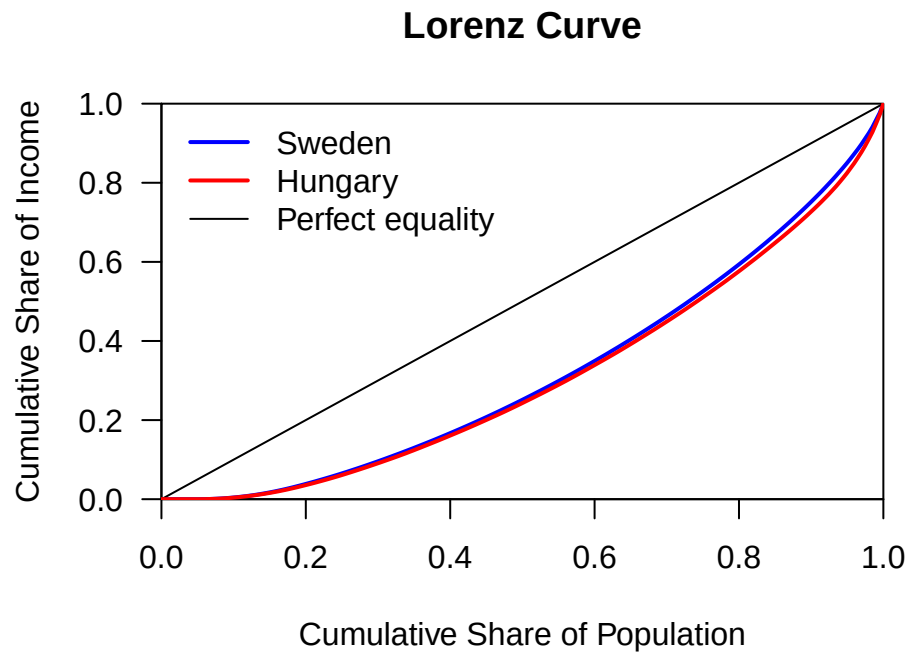


Figure 1: Lorenz curves for Sweden and Hungary

As shown in Figure 1, the Lorenz curves for Sweden and Hungary indicate that Sweden’s curve is closer to the perfect equality line than Hungary’s, implying a more equal distribution in Sweden.

Table 1: Gini coefficients for Sweden and Hungary

Country	Gini - Ineq	Gini - formula
Sweden	0.364	0.366
Hungary	0.383	0.385

As reported in Table 1, the Gini coefficient calculated by *Ineq* package of Sweden is 0.364, slightly lower than Hungary’s (0.383). This indicates that Sweden is more income-equal than Hungary.

However, the Gini coefficients computed via the formula and via the *Ineq* package differ slightly. The formula-based calculation uses the sample covariance, which tends to be a bit larger, whereas the *Ineq* package uses the population covariance, applying an $(n - 1)/n$ adjustment to the formula-based result.

Part 2

I pair Sweden with Hungary because Hungary is an interesting outlier among European countries—sometimes characterized as an “electoral autocracy,”¹ suggesting distinctive institutional features. This contrast offers a useful case for assessing how intergenerational income elasticity differs across institutional contexts.

Based on $y_i = \alpha + \beta x + \delta z + \varepsilon$, I then decompose the variance of y into components of x , z , and their covariance.

Table 2: Variance decomposition results (Sweden)

Component	Value	Share of Total
Household income	6.477	100.0%
Years of education	0.545	8.4%
Father’s years of education	0.057	0.9%
Covariance term	0.126	1.9%

Calibration of R^2 : 0.112

¹<https://www.europarl.europa.eu/news/en/press-room/20220909IPR40137>

As shown in Table 2 and Table 3, the variance decompositions of income inequality in Sweden and Hungary differ markedly. In Sweden, years of education accounts for 8.4% of the variance in household income, father’s education accounts for 0.9%, and the covariance term accounts for 1.9%. In Hungary, the respective shares are 4.1%, 4.7%, and 4.0%. These patterns indicate that individual’s years of education is the dominant observed factor in Sweden, whereas in Hungary father’s years of education and the covariance between individual’s education and father’s education account for a large share.

Table 3: Variance decomposition results (Hungary)

Component	Value	Share of Total
Household income	7.555	100.0%
Years of education	0.313	4.1%
Father’s years of education	0.352	4.7%
Covariance term	0.300	4.0%
Calibration of R^2 : 0.128		

The decomposition suggests that intergenerational income elasticity partially caused by father’s education and its covariance with individual’s education are more salient in Hungary. Institutional differences could underlie these contrasts. In other words, Sweden and Hungary differ in their intergenerational income inequality mechanisms.

Taken together, the Lorenz curves and Gini coefficients align with the variance decomposition: Sweden’s lower overall inequality coincides with a larger role for individual’s education, whereas the pattern in Hungary highlights father’s education.